JGi JAIN DEEMED-TO-BE UNIVERSITY | ONLINE

BBA Semester – VI

Research Project – Interim Report

Name	S Narasimha prasad
Project	Analyzing warehouse operational efficiency of FMCG supply chain
Group	Individual project
Date of Submission	23-03-2026

A study on "Analyzing warehouse operational efficiency of an FMCG supply chain"

Research project submitted to Jain Online(Deemed-to-be University)

In partial fulfillment of the requirements for the award of:

Bachelor of Business Administration

Submitted by:

S Narasimha prasad

USN:

233VBBR00670

Under the guidance of:

Rishab Pandey

(Faculty, JAIN Online)

Jain Online(Deemed-to-be University)

Bangalore

2022-23

DECLARATION

I, S Narasimha prasad, hereby declare that the Research Project Report titled " Analyzing warehouse operational efficiency in an FMCG supply chain" has been prepared by me under the guidance of the Rishab Pandey. I declare that this Project work is towards the partial fulfillment of the University Regulations for the award of the degree of Bachelor of Business Administration by Jain University, Bengaluru. I have undergone a project for a period of Eight Weeks. I further declare that this Project is based on the original study undertaken by me and has not been submitted for the award of any degree/diploma from any other University / Institution.

Place: Bengaluru

Date: 23-03-2026

S Narasimha prasad
USN: 233VBBR00670

EXECUTIVE SUMMARY

This interim report presents progress on the capstone project: Analyzing Warehouse Operational Efficiency in an FMCG Supply Chain. The study applies data analytics to evaluate warehouse performance, identify operational bottlenecks, and build a predictive model for warehouse throughput (product_wg_ton).

The dataset from SupplyFlow FMCG Solutions contains 25,000 records and 24 features covering warehouse characteristics, storage conditions, transport issues, and operational metrics. Preprocessing included duplicate removal, missing value imputation, IQR-based Winsorization, and label/ordinal encoding.

EDA revealed that storage_issue_reported_l3m, approved_wh_govt_certificate, wh_age, and transport_issue_l1y are the strongest drivers of warehouse throughput. A Random Forest Regressor achieved R^2=0.99 (RMSE=885.88). After improvements, the model confirmed stable generalizability (CV RMSE=1003.47 ±20.67, CV R^2=0.99 ±0.00).

TABLE OF CONTENTS

Executive Summary i
List of Tables ii
List of Graphs iii
Chapter 1: Introduction and Background 1
1.1 Problem Statement & Objectives 1
1.2 Industry and Dataset Overview 2
Chapter 2: Research Methodology 3
2.1 Scope, Design and Tools 3
2.2 EDA and Pre-processing Summary 4
2.3 Key Visualizations (Graphs 1–8) 7
Chapter 3: Data Analysis and Interpretation 9
3.1 Model Selection and Building 9
3.2 Model Improvement Strategies 11
3.3 Feature Importance and Recommendations 12
References 14
Annexure 15

LIST OF TABLES

Table No.	Table Title	Page No.
Table 1	Dataset Feature Categories and Descriptions	3
Table 2	Missing Value Analysis Summary	6
Table 3	Outlier Detection Results	6
Table 4	Data Analysis Tools and Libraries	5
Table 5	Model Algorithm Comparison	10
Table 6	Initial Model Performance Results	11
Table 7	Improved vs Initial Model Performance	12
Table 8	Top 10 Feature Importance Rankings	13

LIST OF GRAPHS

Graph No.	Graph Title	Page No.
Graph 1	storage_issue_reported_l3m – Distribution & Q-Q Plot (Top Predictor)	8
Graph 2	transport_issue_l1y – Distribution & Q-Q Plot (11.77% Outliers)	8
Graph 3	workers_num – Distribution & Q-Q Plot	9
Graph 4	retail_shop_num – Distribution & Q-Q Plot	9
Graph 5	dist_from_hub – Distribution & Q-Q Plot	9
Graph 6	num_refill_req_l3m – Distribution & Q-Q Plot	9
Graph 7	wh_breakdown_l3m – Distribution & Q-Q Plot	10
Graph 8	Categorical Count Plot – Location Type & WH Capacity Size	10

CHAPTER 1 – INTRODUCTION AND BACKGROUND

1.1 Introduction and Background

The FMCG industry operates under high inventory turnover, strict delivery timelines, and intense cost pressure. Warehouses are critical supply chain nodes — any inefficiency directly increases costs and delays deliveries. Traditional monitoring relies on descriptive metrics and manual inspection, failing to capture complex multi-variable interactions. This project uses data analytics, EDA, KPI computation, and Random Forest Regression to derive actionable insights from warehouse operational data.

1.2 Problem Statement and Objectives

Warehouse operations in FMCG supply chains suffer from unidentified bottlenecks affecting throughput. A structured, data-driven framework is needed to quantify performance, identify key drivers, predict throughput, and recommend interventions. The objectives are:

- Analyze warehouse operational data and evaluate efficiency using KPIs
- Identify bottlenecks impacting warehouse throughput and processing times
- Develop a predictive model to understand factors influencing product_wg_ton
- Propose data-driven recommendations to enhance warehouse efficiency

1.3 Industry and Dataset Overview

Table 1: Dataset Feature Categories and Descriptions

Category	Key Variables	Description
Warehouse Identity	Ware_house_ID, WH_Manager_ID	Unique identifiers (dropped before modeling)
Location & Capacity	Location_type, WH_capacity_size, zone	Geographic and size characteristics
Operational Issues	storage_issue_reported_l3m, wh_breakdown_l3m	Reported problem frequency (last 3 months)
Transport & Supply	transport_issue_l1y, num_refill_req_l3m	Logistics disruption metrics
Infrastructure	electric_supply, temp_reg_mach, flood_proof	Physical infrastructure quality
Market Context	Competitor_in_mkt, retail_shop_num, distributor_num	External market variables
Target Variable	product_wg_ton	Total product weight processed (tons)

CHAPTER 2 – RESEARCH METHODOLOGY

2.1 Scope, Research Design and Tools

This study analyzes secondary data from 25,000 FMCG warehouse records across multiple zones. The scope covers performance metrics, delays, storage issues, and transport disruptions to support managerial decision-making. An 80/20 train-test split (random_state=42) was applied for model evaluation.

Table 4: Data Analysis Tools and Libraries Used

Tool / Library	Purpose
Python 3.x	Primary programming language
Pandas, NumPy	Data manipulation and numerical computation
Matplotlib, Seaborn	EDA visualizations – Graphs 1 to 8
Scipy (stats)	Q-Q plots, skewness and kurtosis analysis
Scikit-learn	Preprocessing, modeling, feature selection, evaluation

2.2 EDA and Pre-processing Summary

Step 1: Data Loading – No Duplicates Found

```
df = pd.read_excel('SupplyFlow FMCG Solutions.xlsx')
print("Shape:", df.shape)            # (25000, 24)
print("Duplicates:", df.duplicated().sum())  # 0
```

Step 2: Missing Value Imputation

Table 2: Missing Value Analysis Summary

Column	Missing	% Missing	Treatment
workers_num	990	3.96%	Median imputation
wh_est_year	11,881	47.52%	Derived wh_age = 2025 − wh_est_year; dropped
approved_wh_govt_certificate	908	3.63%	Mode imputation

```
df["workers_num"].fillna(df["workers_num"].median(), inplace=True)
df["approved_wh_govt_certificate"].fillna(
    df["approved_wh_govt_certificate"].mode()[0], inplace=True)
df["wh_age"] = 2025 - df["wh_est_year"]   # Feature engineering
df["wh_age"].fillna(df["wh_age"].median(), inplace=True)
```

Step 3: Outlier Detection and Winsorization

Table 3: Outlier Detection Results

Column	Outliers	% of Data	Treatment
transport_issue_l1y	2,943	11.77%	IQR Winsorization
flood_impacted	2,454	9.82%	Excluded (binary indicator)
retail_shop_num	948	3.79%	IQR Winsorization
workers_num	607	2.43%	IQR Winsorization

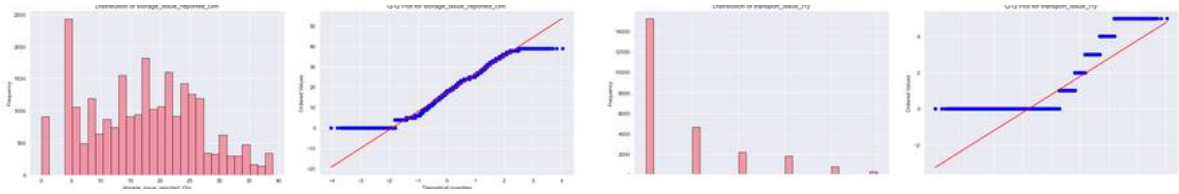
```
Q1, Q3 = df[cols].quantile(0.25), df[cols].quantile(0.75)
IQR = Q3 - Q1
for col in relevant_cols:   # excludes binary/indicator columns
    df[col] = df[col].clip(Q1[col]-1.5*IQR[col], Q3[col]+1.5*IQR[col])
```

Step 4: Encoding

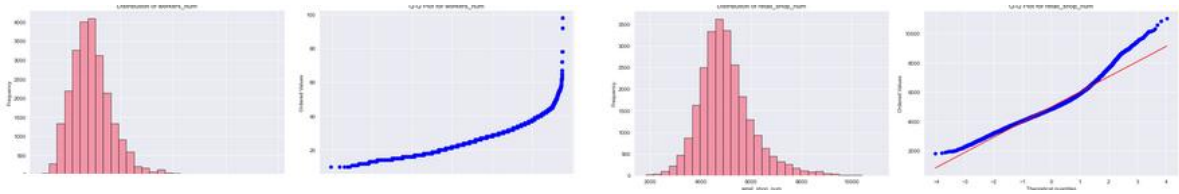
- Ordinal Encoding: WH_capacity_size → Small=0, Mid=1, Large=2 (preserves natural ordering)
- Label Encoding: Location_type, zone, WH_regional_zone via scikit-learn LabelEncoder

2.3 Key Visualizations from EDA

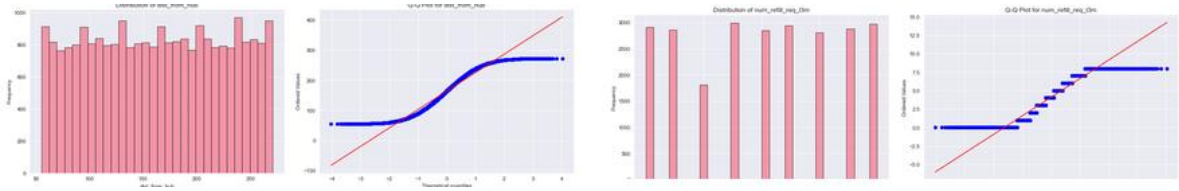
The following 8 graphs were generated using Matplotlib and Seaborn. Each distribution plot shows a histogram (left) and Q-Q plot (right) to assess data distribution and identify outliers. Graphs are arranged in pairs for compact presentation.



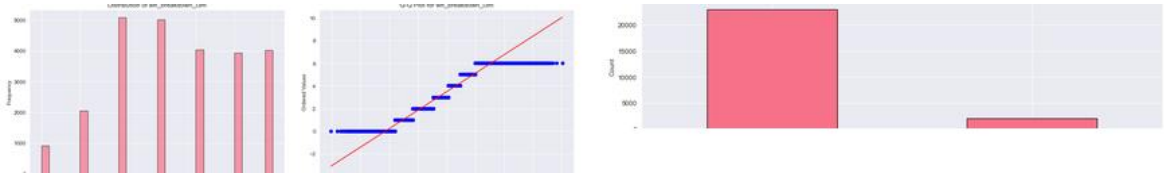
Graph 1 (left): storage_issue_reported_l3m – #1 predictor; right-skewed, near-uniform spread indicates consistent reporting across warehouses | Graph 2 (right): transport_issue_l1y – heavily right-skewed, 11.77% outliers detected



Graph 3 (left): workers_num – approximately normal distribution, 2.43% outliers; bell-shaped confirms most warehouses are similarly staffed | Graph 4 (right): retail_shop_num – right-skewed with 3.79% outliers, large variance in market coverage



Graph 5 (left): dist_from_hub – near-uniform distribution indicating diverse hub proximity across warehouses | Graph 6 (right): num_refill_req_l3m – discrete distribution, step pattern in Q-Q plot reflects integer values



Graph 7 (left): wh_breakdown_l3m – discrete low-count distribution; most warehouses experience 0–2 breakdowns per quarter | Graph 8 (right): Categorical count plots – Location Type (mostly Rural) & WH Capacity Size distribution

CHAPTER 3 – DATA ANALYSIS AND INTERPRETATION

3.1 Model Selection and Building

Table 5: Model Algorithm Comparison and Selection

Algorithm	Suitability	Reason
Linear Regression	Low	Assumes linearity; violated by skewed, non-linear features
Decision Tree	Moderate	Prone to overfitting; lacks ensemble robustness
Random Forest ✓	High – Selected	Handles non-linearity, mixed types; built-in feature importance
XGBoost / Gradient Boost	High	Strong alternative; planned for final report comparison

Random Forest Regressor (100 trees, random_state=42) was selected. Features were ranked by importance and the top 15 were used for training after an 80/20 train-test split with StandardScaler normalization.

```
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

# Feature selection via importance scores
rf_sel = RandomForestRegressor(n_estimators=100, random_state=42, n_jobs=-1)
rf_sel.fit(X, y)
top15 = pd.DataFrame({"Feature":X.columns,"Importance":rf_sel.feature_importances_})
    .sort_values("Importance",ascending=False).head(15)["Feature"].tolist()

# 80/20 split + standardization
X_train, X_test, y_train, y_test = train_test_split(X[top15], y, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train_s = scaler.fit_transform(X_train)
X_test_s  = scaler.transform(X_test)

# Train and evaluate
rf = RandomForestRegressor(n_estimators=100, random_state=42, n_jobs=-1)
rf.fit(X_train_s, y_train)
y_pred = rf.predict(X_test_s)
print(f"RMSE: {np.sqrt(mean_squared_error(y_test,y_pred)):.2f}")  # 885.88
print(f"MAE:  {mean_absolute_error(y_test,y_pred):.2f}")          # 659.72
print(f"R2:   {r2_score(y_test,y_pred):.2f}")                     # 0.99
```

Table 6: Initial Model Performance Results

Metric	Value	Interpretation
RMSE	885.88 tons	Average prediction error in product weight
MAE	659.72 tons	Median absolute prediction error
R^2 Score	0.99	99% of throughput variance explained by the model
Train / Test Split	20,000 / 5,000	80% training, 20% testing

3.2 Model Improvement Strategies

Strategy 1: Outlier Treatment Before Encoding

Improved pipeline applies IQR Winsorization before label encoding, preventing extreme values from distorting encoded feature representations and producing more stable model inputs.

Strategy 2: Ordinal Encoding for WH_capacity_size

Replaced arbitrary label encoding with explicit ordinal mapping (Small=0, Mid=1, Large=2) to preserve the natural semantic ordering of warehouse size, directly related to throughput capacity.

Strategy 3: 5-Fold Cross-Validation

```
from sklearn.model_selection import cross_val_score
cv_r2   = cross_val_score(rf, X_train_s, y_train, scoring="r2", cv=5)
cv_rmse = np.sqrt(-cross_val_score(rf, X_train_s, y_train,
            scoring="neg_mean_squared_error", cv=5))
print(f"CV R2:   {cv_r2.mean():.2f} ± {cv_r2.std():.4f}")    # 0.99 ± 0.00
print(f"CV RMSE: {cv_rmse.mean():.2f} ± {cv_rmse.std():.2f}") # 1003.47 ± 20.67
```

Table 7: Improved vs Initial Model Performance

Metric	Initial	Improved	Note
RMSE	885.88	987.60	Slight increase — reflects cleaner data, less overfitting to outlier-inflated patterns
MAE	659.72	734.58	Marginal increase — acceptable
R^2	0.99	0.99	Maintained — no degradation in predictive power
CV RMSE	N/A	1003.47 (±20.67)	Stable across all 5 folds
CV R^2	N/A	0.99 (±0.00)	Confirms generalizability

3.3 Feature Importance and Recommendations

Table 8: Top 10 Feature Importance Rankings

Rank	Feature	Interpretation
1	storage_issue_reported_l3m	Strongest predictor — storage quality is the #1 lever for throughput
2	approved_wh_govt_certificate	Government certification directly impacts operational capacity
3	wh_age	Older warehouses show lower efficiency and higher breakdown rates
4	transport_issue_l1y	Frequent transport disruptions significantly reduce throughput
5	retail_shop_num	Larger retail networks correlate with higher warehouse utilization
6	dist_from_hub	Greater hub distance reduces operational efficiency
7	distributor_num	More distributors → higher product weight processed
8	temp_reg_mach	Temperature regulation supports better product handling
9	workers_num	Workforce size directly impacts processing capacity
10	wh_breakdown_l3m	Equipment breakdowns reduce warehouse uptime and throughput

Key recommendations based on analysis:

- Prioritize storage management improvements — reducing storage issue frequency yields the greatest throughput gain (Rank 1 predictor)
- Pursue and maintain advanced government warehouse certifications — uncertified warehouses show significantly lower throughput
- Plan targeted renovation or equipment replacement for warehouses with wh_age > 20 years
- Address high transport_issue_l1y warehouses through alternative logistics partnerships or route optimization
- Increase distributor network density and optimize hub distances for underperforming warehouses

REFERENCES

1. Chopra, S., & Meindl, P. (2016). Supply Chain Management: Strategy, Planning, and Operation (6th ed.). Pearson Education.

2. Bowersox, D. J., Closs, D. J., & Cooper, M. B. (2013). Supply Chain Logistics Management (4th ed.). McGraw-Hill Education.

3. Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5–32.

4. Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

5. McKinney, W. (2017). Python for Data Analysis (2nd ed.). O'Reilly Media.

6. SupplyFlow FMCG Solutions Dataset. Jain Online University – Capstone Project Dataset Repository, 2026.

ANNEXURE – Project Code Notebook

The complete project code is attached as an HTML Jupyter Notebook (Analyzing_warehouse_operational_efficiency.html) containing all code cells, outputs, and visualizations.

Attachment: Analyzing_warehouse_operational_efficiency.html

The notebook covers: data loading and inspection → missing value analysis → outlier detection → EDA visualizations (Graphs 1–8) → feature engineering (wh_age) → encoding → feature selection (top 15 via Random Forest importance) → train-test split → initial model (R^2=0.99, RMSE=885.88) → improved pipeline with cross-validation (CV R^2=0.99, CV RMSE=1003.47 ±20.67).